Minesweeper Test Cases

Passed Tests

Feature	Input / Action	Expected Result	Actual Result	Pass/Fail
Difficulty selection	Click Beginner, Intermediate, or Expert	Game starts with the selected difficulty and correct board size/mine count	Game starts with correct difficulty	Pass
Reveal cell	Left-click a covered cell	Cell is uncovered and displays the number of adjacent mines	Cell uncovers correctly	Pass
First-click safety	Left-click any cell as the first move	First clicked cell does not contain a mine	First cell is safe	Pass
Place flag	Right-click a covered cell	Flag is placed and remaining flag count decreases by 1	Flag is placed and count decreases	Pass
Remove flag	Right-click a flagged cell	Flag is removed and	Flag is removed and	Pass

		remaining flag count increases by 1	count increases	
Reveal flagged cell	Left-click a flagged cell	Flagged cell remains covered	Flagged cell remains covered	Pass
Mine reveal	Left-click a mine after the first move	Game ends and loss state is displayed	Game ends with loss state	Pass
Win condition	Uncover all non-mine cells	Game ends and win state is displayed	Game ends with win state	Pass
Restart game	Press “R” key during a game	Game restarts using the current difficulty	Game restarts correctly	Pass
Return to menu	Press “Esc” during a game	Game returns to difficulty selection menu	Difficulty menu appears	Pass
Timer start	Reveal the first cell	Timer begins when the first cell is revealed	Timer begins correctly	Pass
Timer stop	Win or lose the game	Timer stops increasing	Timer freezes correctly	Pass

		when game ends		
Board interaction after game over	Click cells after winning or losing	Board no longer accepts reveal or flag actions	Board does not accept additional actions	Pass

AI Solvers

Feature	Input / Action	Expected Result	Actual Result	Pass/Fail
Easy AI Solver	Run Easy AI solver (Interactive and Auto) across Beginner, Intermediate, and Expert difficulties	Solver executes moves correctly according to Easy logic across all difficulties	Easy AI solver functions as expected	Pass
Medium AI Solver	Run Medium AI solver (Interactive and Auto) across Beginner, Intermediate, and Expert difficulties	Solver executes moves correctly according to Medium logic across all difficulties	Medium AI solver functions as expected	Pass

Hard AI Solver	Run Hard AI solver (Interactive and Auto) across Beginner, Intermediate, and Expert difficulties	Solver executes moves correctly according to Hard logic across all difficulties	Hard AI solver functions as expected	Pass

Additional Features

Feature	Input / Action	Expected Result	Actual Result	Pass/Fail
Stats screen tracking	View stats screen after completing games	Tracks wins, losses, win streak, and best time correctly	Stats update and display accurately	Pass
Sound effects	Trigger win and loss game states	Sound effects play appropriately on win/loss	Win/loss sound effects play as expected	Pass
Hint system	Request a hint during gameplay	A safe cell is revealed or highlighted	Hint reveals a safe cell correctly	Pass